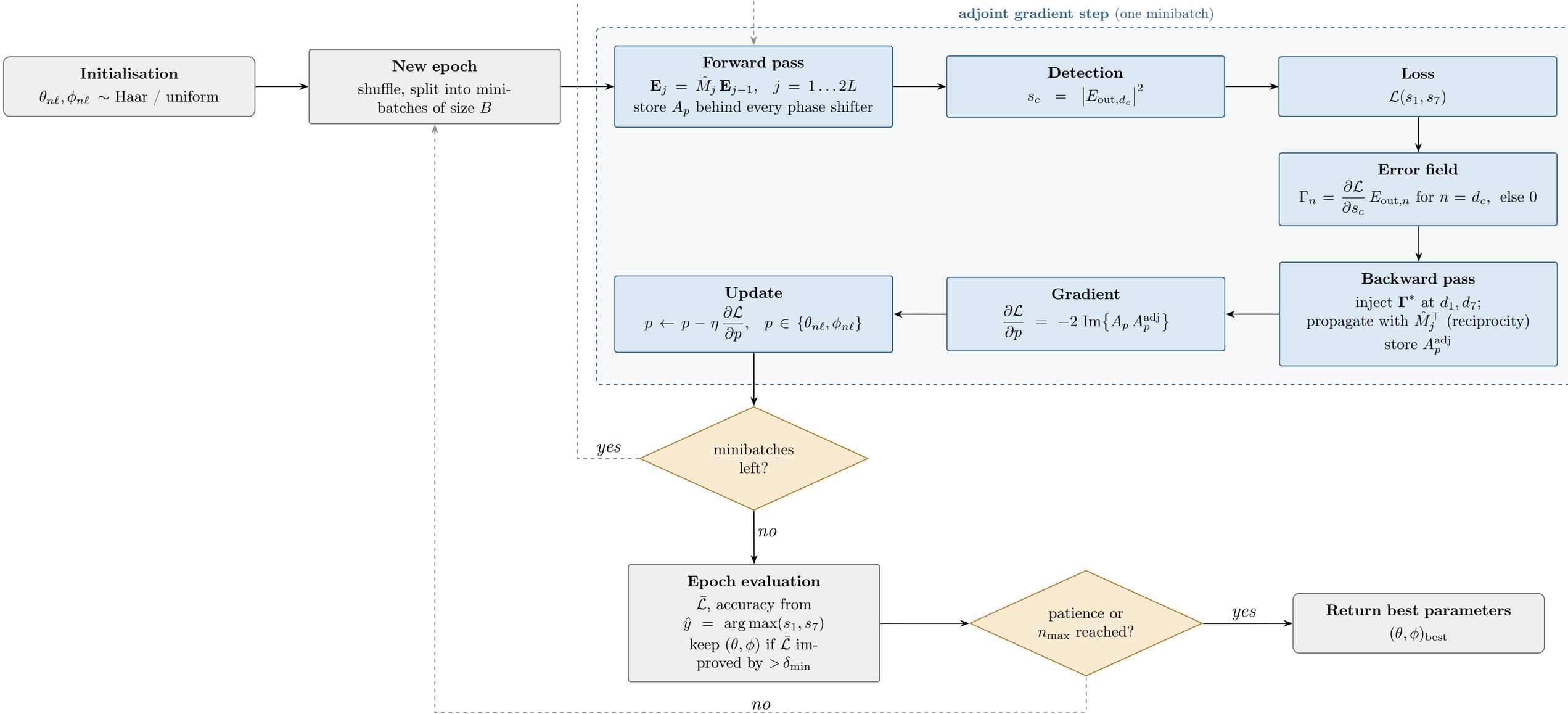

adjoint gradient step (one minibatch)
Initialisation
$\theta_{n\ell}, \phi_{n\ell} \sim$ Haar / uniform
New epoch
shuffle, split into mini-batches of size B
Forward pass
$\mathbf{E}_j = \hat{M}_j \mathbf{E}_{j-1}, \quad j = 1 \dots 2L$
store A_p behind every phase shifter
Detection
$s_c = \left|E_{\text{out},d_c}\right|^2$
Loss
$\mathcal{L}(s_1, s_7)$
Error field
$\Gamma_n = \frac{\partial \mathcal{L}}{\partial s_c} E_{\text{out},n}$ for $n = d_c$, else 0
Backward pass
inject $\mathbf{\Gamma}^*$ at d_1, d_7;
propagate with $\hat{M}_j^\top$ (reciprocity)
store A_p^{adj}
Gradient
$\frac{\partial \mathcal{L}}{\partial p} = -2\,\text{Im}\{A_p\, A_p^{\text{adj}}\}$
Update
$p \leftarrow p - \eta \frac{\partial \mathcal{L}}{\partial p}, \quad p \in \{\theta_{n\ell}, \phi_{n\ell}\}$
minibatches left?
yes
no
Epoch evaluation
$\bar{\mathcal{L}}$, accuracy from
$\hat{y} = \arg\max(s_1, s_7)$
keep (θ, ϕ) if $\bar{\mathcal{L}}$ improved by $> \delta_{\min}$
patience or $n_{\max}$ reached?
yes
no
Return best parameters
$(\theta, \phi)_{\text{best}}$